Assessing Spatial Information in Decomposition-Assisted Ridge–KANInformer
A Seasonal Brentwood Wind-Speed Forecasting Study

Abstract

A single-station forecasting system cannot directly exploit measurements from other stations, motivating the addition of spatial context to a decomposition-assisted Ridge–KANInformer model. This study compares three implementations on new California CIMIS Brentwood data: L fixes seasonal VMD mode counts and selects alpha by Ridge validation prediction error; L.1 selects K by reconstruction error for each alpha and then selects alpha by validation prediction error; Spatial L.1 adds a fixed weighted graph aggregation layer to the neural residual pathway while retaining Brentwood-only Ridge prediction and neural supervision. Thirteen other stations constitute the candidate context pool, subject to seasonal data eligibility. Four seasonal models predict one, two and three hours ahead. Among 36 reported season–horizon metric comparisons, L.1 achieves 20 minimum errors, Spatial L.1 achieves 10 and L achieves 6. Relative to L.1, the spatial extension improves 13 metrics and worsens 23. Equal-weight case means show slightly lower MAE but slightly higher RMSE and MAPE for Spatial L.1. The results support selective rather than uniform spatial benefit. Full-season cleaning and decomposition remain an explicit retrospective leakage protocol. Numerical evidence consists of reported aggregate scores; prediction-level verification, repeated-seed uncertainty and operational forecasting performance remain unestablished.

Keywords: wind-speed forecasting; VMD; empirical wavelets; spatial graph; KANInformer; Ridge residual learning; Brentwood.

1 Introduction

1.1 Background and motivation

The earlier single-station implementation represented Brentwood wind speed through decomposition components, selected weather measurements, a Ridge forecast and a nonlinear KANInformer residual correction. Its temporal features could represent local oscillations but could not directly observe another station at the forecast origin. Availability of additional station records motivated the question: can their historical information improve Brentwood forecasts while the local prediction architecture is retained?

Spatial dependence is a plausible source of information, not a guarantee of improvement. Stations can share atmospheric influences but differ in terrain exposure, local circulation, data quality and timing. Geographic proximity supplies a connection prior without establishing predictive relevance. Redundant or mismatched information may add little beyond a strong local predictor and can worsen a learned correction.

The experiment therefore reruns both local baselines on the new dataset before evaluating a spatial extension. Historical Brentwood errors from another period are not treated as interchangeable test scores. The target remains Brentwood; this study does not jointly forecast all network nodes or interpolate wind at unobserved locations.

1.2 Literature review of wind speed forecasting

1.2.1 Wind speed forecasting research

The reference study by Leng et al. [1] combines improved dual decomposition with KANInformer for short-term wind-speed prediction. The present work develops a target-station spatial extension of its local adaptation. Wind speed, not electrical power, is evaluated. No turbine conversion, wind-farm energy output or economic dispatch study is available, so error reductions are not restated as measured power improvements.

A useful experimental distinction is between additional information and additional supervision. Other stations supply spatial context here, but no other station receives a final neural forecast head or target loss. The comparison consequently examines whether context helps a retained local task rather than whether a larger multitask dataset helps a new task.

1.2.2 Signal decomposition methods

VMD estimates narrow-band signal components [2], whereas EWT builds an adaptive spectral filter bank [3]. The retained workflow applies VMD, calculates sample entropy, aggregates irregular modes and applies EWT to that aggregate. Its final feature bank retains all VMD modes alongside EWT components and raw wind speed.

Reconstruction fidelity and prediction usefulness are different objectives. The representation that most accurately reconstructs a seasonal series need not be the representation best suited to forecasting. L and L.1 explicitly separate these objectives in parameter selection; the spatial extension then adds station context to L.1.

1.2.3 Classification of wind speed forecasting models

The prediction system combines a linear statistical model with a neural temporal model. KANs use learnable univariate functions in network connections [4]; Informer contributes efficient temporal attention and a generative prediction design [5]. Their retained local combination is specified in Section 2.2.

The spatial addition follows neighborhood aggregation ideas associated with graph convolution [6], but uses a custom fixed-weight, target-row operation. It is not an exact original GCN layer, graph attention network, dynamic graph model or exact reproduction of a published spatiotemporal benchmark. The separate advanced-model runners in the wider project are outside this paper's numerical comparison.

1.3 Implementation extensions and contributions

The idea developed from a prediction-guided local baseline L, through hybrid-selection baseline L.1, to Spatial L.1 after other station records became available. No Spatial L experiment is included. The main contributions are:

1. Consolidated definitions of the three architectures and their VMD selection objectives.
2. A target-only spatial adapter supporting unequal station feature widths while retaining the KANInformer temporal backbone.
3. A seasonal new-data comparison of reported Brentwood errors, per-metric winners and relative spatial changes.
4. An account of selective gains, missing neighbor measurements, retrospective leakage and the limits of aggregate-score evidence.

These are implementation and comparative contributions, not a proof of universal spatial benefit or a new VMD optimization algorithm.

2 Methodology

2.1 Improved dual decomposition

Station 0 denotes Brentwood, s denotes season, and $\mathcal{E}_s$ is its eligible station set, including Brentwood. A seven-hour history predicts $H = 3$ future observations:

$$X_{j,o} = [x_{j,o-7}, \ldots, x_{j,o-1}], \quad \mathbf{Y}_{0,o} = (y_{0,o}, y_{0,o+1}, y_{0,o+2})^\top. \tag{1}$$

The index o is the first predicted observation; the forecast-origin timestamp is $o-1$. All stations share this index convention.

2.1.1 Variational Mode Decomposition

VMD estimates modes u_k and center frequencies ω_k through

$$\min_{\{u_k\},\{\omega_k\}} \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{\mathrm{i}}{\pi t} \right) * u_k(t) \right] e^{-\mathrm{i}\omega_k t} \right\|_2^2, \quad \sum_{k=1}^{K} u_k(t) = y(t). \tag{2}$$

The copied package call uses penalty alpha, multiplier step $\tau = 0$, uniform initialization, tolerance 10^{-7} and the package's 500-iteration limit. Modes are ordered by final center frequency. Odd-length signals are edge-padded once and cropped afterward. Finite reconstruction residual is measured rather than assumed absent.

The selection diagnostic is

$$R_{j,s}(a,k) = \frac{1}{N_s} \sum_{t=0}^{N_s-1} \frac{\left| y_{j,t} - \sum_{\ell=1}^{k} u_{j,\ell,t}^{(a,k)} \right|}{\max(|y_{j,t}|, 10^{-12})}. \tag{3}$$

RER in percent is $100R_{j,s}$. This is a mean absolute relative residual, not a squared energy ratio. Near-zero wind can dominate it; a low residual does not demonstrate physical separation of atmospheric processes.

L: prediction-guided alpha at fixed K. The retained seasonal counts are

$$(K^{\mathrm{L}}_{\mathrm{spring}}, K^{\mathrm{L}}_{\mathrm{summer}}, K^{\mathrm{L}}_{\mathrm{autumn}}, K^{\mathrm{L}}_{\mathrm{winter}}) = (11, 10, 11, 10). \tag{4}$$

For $a \in \mathcal{A} = \{25, 100, 500, 2000\}$, a Ridge probe fits training windows from the candidate feature bank:

$$(W_{a,k}, b_{a,k}) = \arg\min_{W,b} \sum_{o \in \mathcal{T}_s} \|\mathbf{Y}_{0,o} - W \,\mathrm{vec}(X_{0,o}^{(a,k)}) - b\|_2^2 + 0.1\|W\|_F^2. \tag{5}$$

Its intercept is not penalized. Validation prediction RMSE is

$$J_s(a,k) = \sqrt{\frac{1}{3|\mathcal{V}_s|} \sum_{o \in \mathcal{V}_s} \|\mathbf{Y}_{0,o} - g_{a,k}(\mathrm{vec}\, X_{0,o}^{(a,k)})\|_2^2}. \tag{6}$$

L chooses $a_s^{\mathrm{L}} = \arg\min_{a \in \mathcal{A}} J_s(a, K_s^{\mathrm{L}})$. Prediction-based VMD therefore means prediction-based parameter selection; the underlying decomposition equation is unchanged.

L.1: hybrid reconstruction and prediction selection. For every alpha, every $k \in \mathcal{K} = \{2, \ldots, 12\}$ is evaluated:

$$k_s^*(a) = \arg\min_{k \in \mathcal{K}} R_{0,s}(a, k), \tag{7}$$

$$a_s^{\mathrm{L.1}} = \arg\min_{a \in \mathcal{A}} J_s(a, k_s^*(a)), \qquad K_s^{\mathrm{L.1}} = k_s^*(a_s^{\mathrm{L.1}}). \tag{8}$$

Exact RER ties prefer smaller K; exact validation ties prefer smaller alpha. The search never stops at the first RER below three percent. Only four reconstruction-selected finalists receive the Ridge prediction comparison, rather than neural training of all 44 candidate pairs. This is neither purely reconstruction-based nor purely prediction-based joint selection.

Spatial L.1: retain target selection and add context. Brentwood uses L.1's selection rule. A same-run selected pair can be reused when data, period and configuration agree; otherwise the same procedure is recomputed. Each eligible neighbor has its own hybrid decomposition. Neighbor prediction probes select auxiliary representations, but final neural supervision remains Brentwood-only.

Table 1: Definitions of the three experiments. S denotes Spatial L.1.

Item	L	L.1	S
Neural target	Brentwood	Brentwood	Brentwood
VMD K	Fixed 11/10/11/10	Minimum RER per alpha	Same target rule as L.1
VMD alpha	Minimum probe RMSE	Minimum probe RMSE after K selection	Same target rule as L.1
Other-station histories	Absent	Absent	Graph context in neural path
Ridge inputs	Own station	Own station	Own station only
Temporal backbone	KANInformer	KANInformer	Same backbone plus spatial adapter

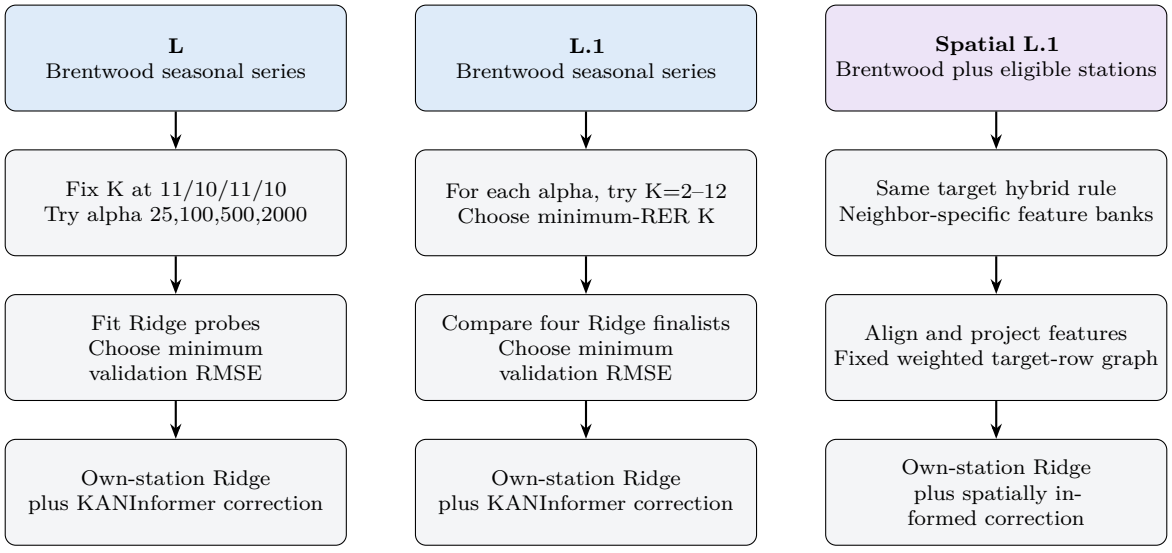


Figure 1: Selection and prediction flows. L to L.1 changes representation selection; L.1 to Spatial L.1 adds context to the neural pathway. No Spatial L experiment is included.

Figure 1 separates two changes that otherwise could be conflated. L versus Spatial L.1 changes both representation selection and spatial input; its entire difference cannot be attributed to graph exchange.

2.1.2 Empirical Wavelet Transform

Sample entropy compares non-self matching templates of lengths m and $m+1$, with maximum-coordinate matching distance:

$$\mathrm{SE}(u; m, r) = -\log\left(\frac{A_{m+1}(r)}{B_m(r)}\right), \qquad m = 2, \quad r = 0.2\ \mathrm{std}(u). \tag{9}$$

The implementation uses equal eligible template starting positions for both lengths. Constant components return zero entropy; zero matching counts produce infinite entropy rather than an invented finite value. The irregular aggregate is

$$q_j(t) = \sum_{k:\mathrm{SE}(u_{j,k})>0.4} u_{j,k}(t). \tag{10}$$

For a nonempty set, EWT creates eight empirical Meyer bands using local spectral maxima, average regularization and completion. Reflection extension and energy-corrected synthesis give

$$e_{j,b}(t) = \mathcal{F}^{-1}\left[\widehat{q}_j(\omega)\frac{|\psi_{j,b}(\omega)|^2}{\sum_\ell |\psi_{j,\ell}(\omega)|^2}\right](t), \tag{11}$$

cropped to the original interval. This equation expresses the implemented synthesis correction, not a replacement for adaptive filter construction. If no mode exceeds the entropy threshold, no EWT aggregate is formed.

2.1.3 VMD-CA-EWT algorithm

Let $m_j(t)$ contain selected weather channels other than WS. Features are

$$x_j(t) = [m_j(t), u_{j,1}(t), \ldots, u_{j,K_j}(t), e_{j,1}(t), \ldots, e_{j,B_j}(t), y_j(t)]. \tag{12}$$

Here $B_j = 8$ when EWT is used, otherwise zero. Every VMD mode remains, including modes in the irregular aggregate. This inherited redundant bank is not a sum of independently predicted components. Training-row standardization gives

$$\widetilde{x}_{j,f}(t) = \frac{x_{j,f}(t) - \mu^{\mathcal{T}}_{j,f}}{\sigma^{\mathcal{T}}_{j,f}}, \tag{13}$$

with the scaler package's constant-feature convention. Stations have separate scalers and potentially different widths F_j. They align by timestamp, not by assuming equal IMF indices have identical meaning.

Full-season cleaning, VMD, entropy grouping and EWT precede the use of chronological partitions for fitting. Training-only scaling and validation-only scoring therefore do not eliminate dependence on future observations. This deliberate retrospective leakage protocol is retained for comparability and is not called operational forecasting.

2.2 KANInformer

2.2.1 Kolmogorov-Arnold Networks

The retained KAN block maps each temporal position through learnable univariate connections. For layer ℓ,

$$z_j^{(\ell+1)} = \sum_i \phi_{ji}^{(\ell)}(z_i^{(\ell)}), \qquad \phi_{ji}(z) = w_{ji}^{\mathrm{base}}\,\mathrm{SiLU}(z) + w_{ji}^{\mathrm{spline}}\sum_b c_{ji,b}B_b(z). \tag{14}$$

The copied implementation uses cubic splines, grid size five and a 64–20–64 KAN block in the specified 64-dimensional temporal representation. The local paper-mode pathway applies the KAN per time position; it does not flatten the entire temporal sequence into a single KAN input. The representation motivation follows KAN [4]; the retained configuration is an implementation choice rather than a theoretical guarantee of greater forecasting accuracy.

2.2.2 Informer

An attention operation is based on

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d})V. \tag{15}$$

The local self-attention implementation samples keys to estimate query sparsity and updates a selected query subset. Encoder blocks combine ProbSparse attention, residual additions, layer normalization and KAN blocks. Between the two encoder blocks, a convolution, batch normalization, ELU and max-pooling distill the sequence from seven to four positions under the default window.

The decoder receives the final four observed feature vectors followed by three zero feature placeholders. It applies causal ProbSparse self-attention, full cross-attention to encoder memory, and a KAN block. A scalar output head at each of the final three positions produces the three residual predictions in one forward pass. This is a direct multihorizon output rather than recursive feeding of future predicted wind values. Informer's original long-sequence motivation [5] does not establish a computational advantage for this short seven-hour window; no speed comparison is supplied here.

2.2.3 Forecasting network combining KAN and Informer

For a station feature width F_0, the original input projection maps $B \times 7 \times F_0$ to $B \times 7 \times 64$. Circular temporal value convolutions and sinusoidal position embeddings precede the encoder and decoder attention stacks. The temporal network returns $B \times 3$ scaled residual forecasts. Spatial L.1 retains this backbone, including its original input dimension F_0, and enriches its input before that projection rather than replacing its encoder or decoder.

Figure 2 makes the architecture-level distinction explicit: adding spatial aggregation does not change the network into a multitask station forecaster. Decoder placeholders contain no future weather observations. Nevertheless, the upstream full-season decomposition creates future dependence before these histories are formed.

2.2.4 Ridge-assisted residual prediction

A separate Ridge model produces the local baseline:

$$g_s(X_{0,o}) = W_s\,\mathrm{vec}(X_{0,o}) + b_s, \tag{16}$$

with the objective in Equation (5) at the selected representation. This final Ridge fit is distinct from candidate-selection probes. For each horizon, the training residual $\mathbf{e}_o = \mathbf{Y}_{0,o} - g_s(X_{0,o})$ is transformed by the original MinMaxScaler:

$$z_{o,h} = \eta_h e_{o,h} + \nu_h, \qquad \widehat{\mathbf{Y}}_{0,o} = g_s(X_{0,o}) + (\widehat{\mathbf{z}}_o - \boldsymbol{\nu}) \oslash \boldsymbol{\eta}. \tag{17}$$

The scaler parameters are fitted on training residuals only. This is not the residual standard-deviation normalization used in a different multistation runner elsewhere in the project.

The neural objective is

$$\mathcal{L}_{\mathrm{train}}(\theta) = \frac{1}{3|\mathcal{T}_s|} \sum_{o \in \mathcal{T}_s} \|f_\theta(\mathcal{X}_o) - \mathbf{z}_o\|_2^2. \tag{18}$$

Ridge coefficients remain fixed during neural optimization. Checkpoint selection uses the combined raw-unit validation MSE:

$$\theta^* = \arg \min_{\theta \text{ at saved epochs}} \frac{1}{3|\mathcal{V}_s|} \sum_{o \in \mathcal{V}_s} \|\widehat{\mathbf{Y}}_{0,o}^{(\theta)} - \mathbf{Y}_{0,o}\|_2^2. \tag{19}$$

The objective contains only Brentwood targets. Predicting other stations' future wind is not an auxiliary neural training task in Spatial L.1.

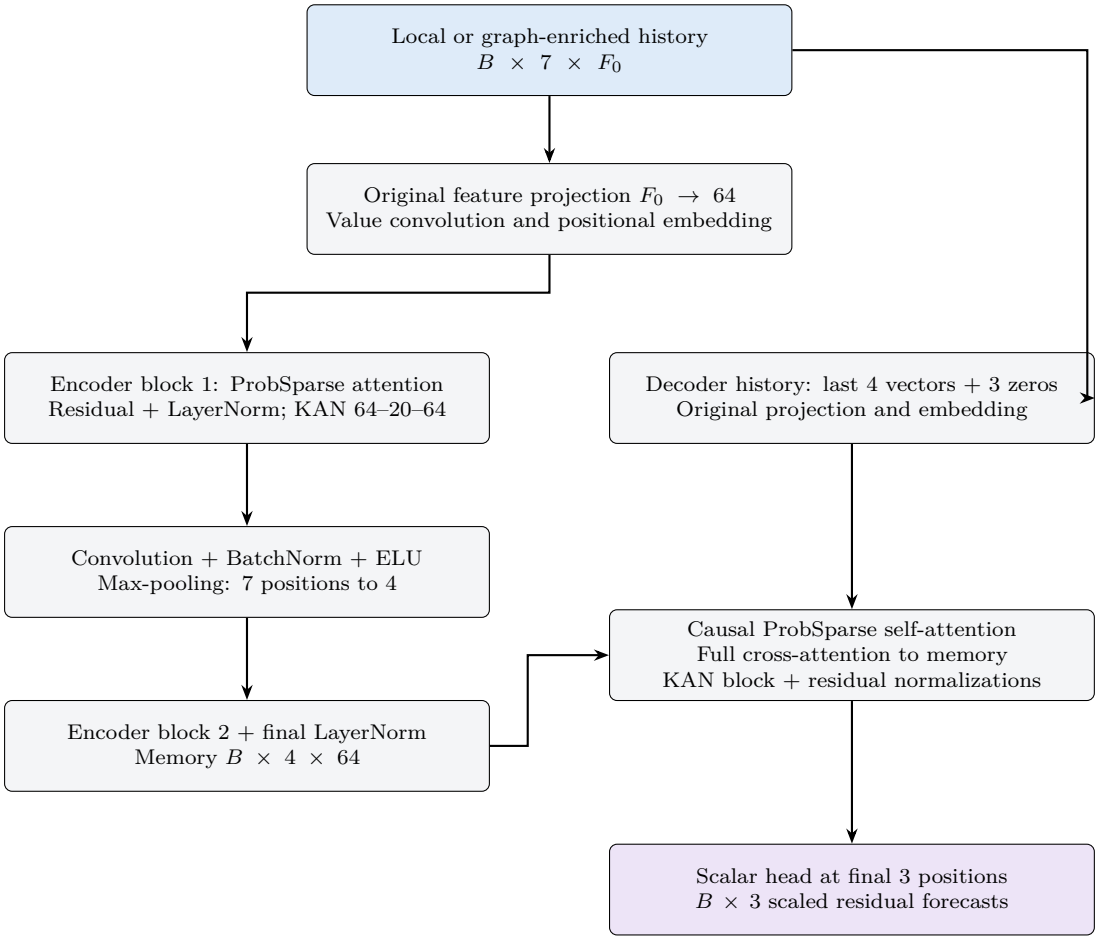


Figure 2: Retained KANInformer temporal backbone. The spatial adapter changes the historical feature input, while the encoder–decoder structure and three-output residual task are retained. Shapes show the documented default configuration.

2.3 Target-specific spatial extension

2.3.1 Seasonal station eligibility and alignment

The candidate network contains Brentwood and thirteen other stations. For every season, the retained cleaner requires at least four finite post-outlier observations for each of the ten meteorological variables before cubic interpolation. A station with an entirely missing variable cannot satisfy that condition. Because cleaning precedes feature selection, the condition also applies to variables not ultimately retained in that season's feature vector.

The execution was reported to exclude unusable neighbor stations and record their reasons, while keeping Brentwood's timeline unchanged. Define the retained set

$$\mathcal{E}_s = \{0\} \cup \{j : \text{station } j \text{ passes the seasonal neighbor cleaning checks}\}. \tag{20}$$

Brentwood is never silently excluded. A failure of its own preparation invalidates the experiment rather than changing the forecast target. Neighbor preparation uses the same calendar interval and the same target origin indices as Brentwood. Incomplete neighbor measurements do not justify dropping different Brentwood evaluation windows.

The reported remote execution change excludes Shafter (station 5) when ET is wholly unavailable and Firebaugh/Telles (station 7) in summer for the same reason. This change is described as a reported execution behavior: the supplied aggregate scores do not provide the complete season-specific exclusion manifests, and the locally retained source snapshot does not contain that exception handler. The final executed version and its manifests must accompany a reproducible submission.

2.3.2 Geographic graph construction

For coordinates in radians, pair distance is

$$d_{ij} = 2R_\oplus \arcsin\left[\sqrt{\sin^2\frac{\varphi_i - \varphi_j}{2} + \cos\varphi_i \cos\varphi_j \sin^2\frac{\lambda_i - \lambda_j}{2}}\right], \quad R_\oplus = 6371.0088 \text{ km}. \tag{21}$$

On eligible nodes, each station selects its three nearest geographic neighbors under the documented default. The binary support is symmetrized:

$$S_{ij} = \mathbb{1}[j \in \mathcal{N}_3(i) \text{ or } i \in \mathcal{N}_3(j)], \quad S_{ii} = 0. \tag{22}$$

A Gaussian distance kernel and self loops produce

$$G_{ij} = S_{ij} \exp[-(d_{ij}/\sigma)^2] \ (i \neq j), \quad G_{ii} = 1, \quad A_{ij} = \frac{G_{ij}}{\sum_\ell G_{i\ell}}, \tag{23}$$

where σ is the median supported off-diagonal distance. Thus rows sum to one; the row-normalized matrix need not be symmetric. Symmetrization can give Brentwood more than three off-diagonal links because another station may nominate Brentwood as its neighbor.

The implementation also permits statistical and combined supports. Statistical support ranks training-row wind correlations, retains nonnegative correlation weights on symmetrized nearest-correlation neighbors, and adds self loops; the combined option equally mixes unnormalized geographic and statistical weights before row normalization. The default geographic specification is documented here, not claimed to be confirmed by an unavailable run adjacency. No final edges are inferred from aggregate accuracy scores.

2.3.3 Unequal feature widths and target-row aggregation

Station-specific K, meteorological lists and entropy outcomes can yield different F_j. The spatial adapter learns a projection for each neighbor:

$$p_{j,t} = P_j \widetilde{x}_{j,t} + b_j \in \mathbb{R}^{F_0} \quad (j \neq 0), \qquad p_{0,t} = \widetilde{x}_{0,t}. \tag{24}$$

Brentwood's own projection is the identity. Packing concatenated histories is a storage operation, not an assumption of common component semantics.

Only the target row of the graph is used:

$$c_{0,t} = \sum_{j \in \mathcal{E}_s} A_{0j} p_{j,t}, \tag{25}$$

$$\widetilde{x}'_{0,t} = \mathrm{LN}(\widetilde{x}_{0,t} + \mathrm{GELU}(W_g c_{0,t} + b_g)) \,. \tag{26}$$

The fixed edges determine who can contribute, while the projections and W_g, b_g learn feature transformations. The layer is GCN-style aggregation with residual addition and normalization; it does not learn individual connection attention coefficients as a GAT would.

For L.1, $\mathcal{X}_o = X_{0,o}$ in Equation (18); for Spatial L.1, $\mathcal{X}_o = X'_{0,o}$. The original temporal network consumes the enriched $7 \times F_0$ history. A station with $A_{0j} = 0$ has no direct contribution through this one-hop target-row operation. The existence of thirteen candidate neighbors does not mean all thirteen affect the forecast. No multi-hop graph stack or physical wind-direction propagation model is implemented.

2.3.4 Preserving the local Ridge pathway

For packed neural inputs $U_o = [X_{0,o}, X_{1,o}, \ldots]$, the spatial-only Ridge input adapter extracts precisely the original Brentwood block:

$$g_s^{\mathrm{S}}(U_o) = W_s \,\mathrm{vec}(X_{0,o}) + b_s. \tag{27}$$

The new context enters the neural correction but never the linear predictor. Source L and L.1 modules remain byte-identical; their baseline model and Ridge classes are not replaced during baseline runs. Only the new spatial experiment supplies the graph adapter and the own-feature extraction adapter to its separately loaded training module. This preserves the local Ridge problem while changing the residual input.

3 The spatial VMD-CA-EWT-Ridge–KANInformer forecasting model

The complete workflow in Figure 3 has a local linear branch and a spatial neural branch. It is executed separately for each season rather than training a single model on disconnected seasonal segments.

1. **Select the new-data period.** Use a complete seasonal cycle with Brentwood as the sole target. Convert source timestamps to the original Date/Hour representation without changing measurements.

2. **Preserve the baseline processing.** Apply full-season three-sigma cleaning and cubic imputation. Construct L using its fixed seasonal K and L.1 using its hybrid selection.

3. **Define chronological partitions.** Set training end $T_s = \lfloor 0.8 N_s \rfloor$, validation end $V_s = \lfloor 0.9 N_s \rfloor$, and reserve the final interval for testing. Use identical Brentwood origins across experiments.

4. **Prepare spatial candidates.** Exclude only unusable neighbors with a recorded reason, prepare their hybrid feature banks, and align their input histories to the unchanged Brentwood windows.

5. **Construct target context.** Form the eligible-node graph, project neighbor features, and aggregate the Brentwood row. Preserve the separate Brentwood feature block for Ridge.

6. **Fit the two-stage predictor.** Fit Ridge on Brentwood training windows. Fit the training residual MinMax transform. Optimize the neural correction using either local or graph-enriched histories.

7. **Choose a checkpoint.** Evaluate the combined raw-unit validation forecast, apply the documented early-stopping rule, and restore the best saved neural weights.

8. **Evaluate Brentwood only.** Export three-horizon test predictions and metrics, compare identical target records, and report per-season changes and winners. Do not optimize graph choices using test scores.

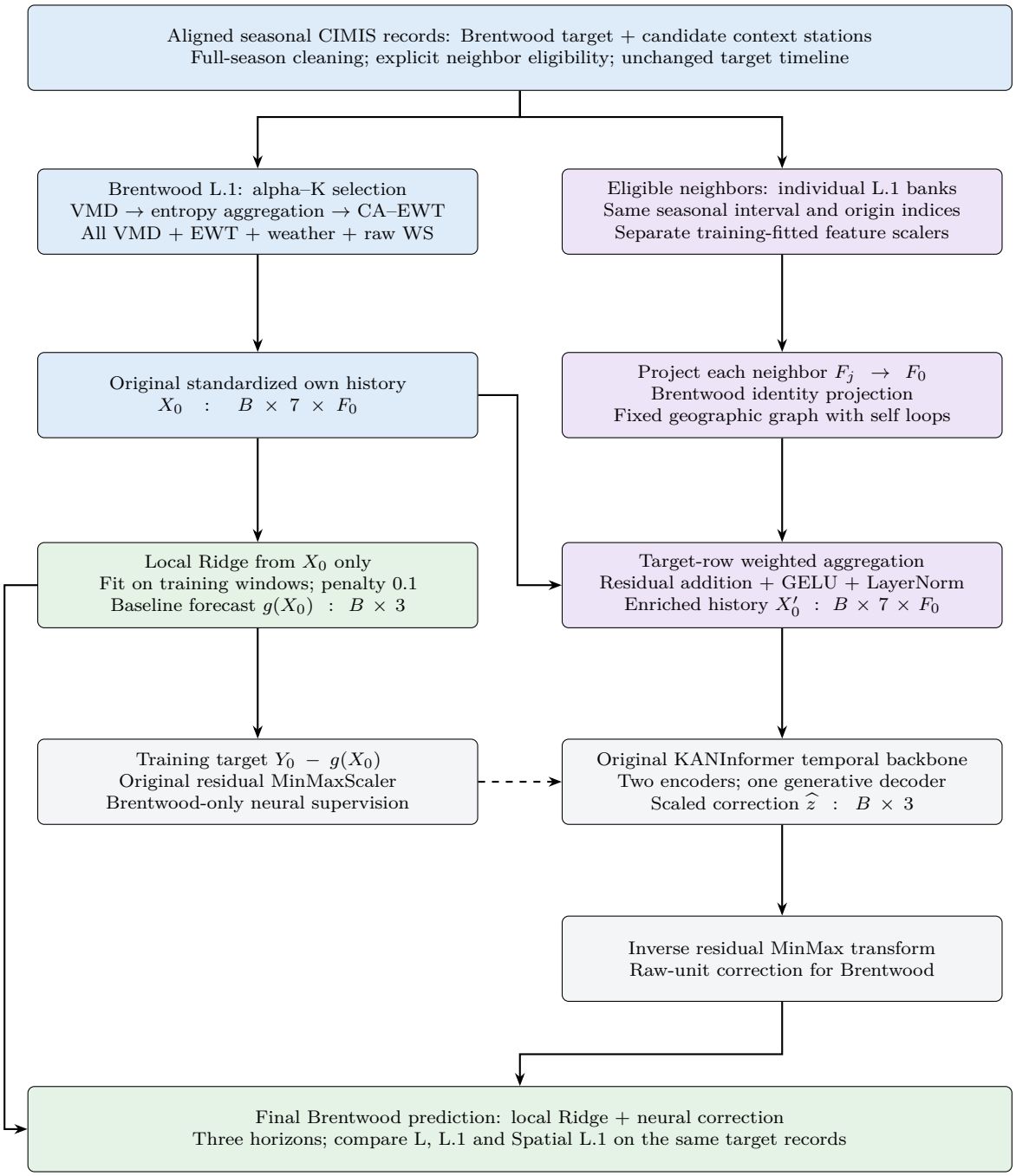


Figure 3: Spatial L.1 architecture. Solid arrows show feature/prediction flow; the dashed arrow shows the supervised residual target. The graph modifies only the neural input. Ridge never receives neighboring station features, and no neighboring neural forecast is required.

Figure 3 also explains why the spatial result is not a neighbor-Ridge baseline: the linear model

remains purely local. An accuracy change can arise from additional station information, learned projections, normalization or altered optimization. A self-only graph adapter control would be required to isolate the benefit of the neighbor signals from the extra adapter capacity itself.

4 Case study

The case study uses California CIMIS station 47, Brentwood, as the forecast target. The source-data files identify fourteen stations in total; the other thirteen supply a candidate spatial context pool. This is the target-specific comparison project, not the separate whole-network model whose output tables average errors across stations.

4.1 Data retrieval and preprocessing

New dataset and source units. The supplied station files cover records beginning in January 2024 and extending into January 2026. The comparison wrappers select a complete 2025 seasonal cycle: winter spans December 2024–February 2025 and the other seasons lie in calendar year 2025. The measurements are from California CIMIS [7]. The inherited coordinates are attributed to the California CIMIS station metadata asset [8]; they are not new field-surveyed positions and no terrain elevation layer is used in the forecasting graph.

Schema conversion retains measurement values. Source Date/Hour fields are reconstructed from fixed PST timestamps; 2400 denotes midnight after the preceding calendar date. Wind is converted from mph to $\mathrm{m\,s^{-1}}$ using 0.44704, and air/dew-point/soil temperature from Fahrenheit to Celsius using $(F - 32)5/9$. The other source units are retained by the original baseline loader, including ET in inches. The separate spatial analysis loader's millimetre conversion is not substituted into this protected-baseline comparison.

Table 2: Candidate station pool from the inherited coordinate asset. These are candidate positions, not a verified list of nonzero edges or retained nodes in the reported run.

ID	Station	County	Latitude	Longitude
2	FivePoints	Fresno	36.3362	-120.1129
5	Shafter	Kern	35.5327	-119.2819
6	Davis	Yolo	38.5358	-121.7764
7	Firebaugh/Telles	Fresno	36.8512	-120.5910
13	Camino	El Dorado	38.7523	-120.7339
15	Stratford	Kings	36.1581	-119.8514
39	Parlier	Fresno	36.5975	-119.5041
47	**Brentwood (target)**	Contra Costa	37.9282	-121.6597
52	San Luis Obispo	San Luis Obispo	35.3054	-120.6618
91	Tulelake FS	Siskiyou	41.9589	-121.4724
105	Westlands	Fresno	36.6340	-120.3818
107	Santa Barbara	Santa Barbara	34.4374	-119.7373
113	King City–Oasis Rd.	Monterey	36.1213	-121.0845
146	Belridge	Kern	35.5058	-119.6911

Original full-season cleaning. For variable v, observations satisfying

$$|v_t - \overline{v}_s| > 3\, s_{v,s} \tag{28}$$

are masked using the full-season mean and sample standard deviation. Remaining missing positions are interpolated with a not-a-knot cubic spline; endpoints can involve spline extrapolation. All ten

variables are processed. No additional causal filling, physical clipping or new QC-code rejection is inserted into the protected baseline functions.

This choice retains the original implementation but has consequences. The preprocessing uses future observations, may replace extreme yet real events, and may evaluate forecasts against imputed or outlier-replaced wind values. The reported errors should therefore be described as errors against the processed target series. A clean comparison of operational forecast skill would require a separate chronological preprocessing policy and scoring against eligible untouched observed targets.

Neighbor-data failures. The reported exclusion of station 5 and the summer exclusion of station 7 avoid attempting a spline where ET has no valid anchors. They do not justify inventing ET or shortening Brentwood's target period. After exclusion the spatial graph should be rebuilt and its station order saved. The exact retained station sets and edges cannot be reconstructed from aggregate metrics alone. A full submission needs the executed exclusion log, final adjacency and any additional exclusions rather than assuming that only the two reported problems occurred.

Seasonal splits and evaluation counts. For seven historical samples and three horizons, training origins satisfy $o+3 \le T_s$, validation origins satisfy $o \ge T_s,\ o+3 \le V_s$, and test origins satisfy $o \ge V_s$. The last allowed origin is $N_s - 3$. Boundary-straddling targets are discarded rather than reassigned. Historical input may include an earlier partition's last observed rows, which are available at the origin.

Table 3: Calendar periods and window counts under the documented default split. Test counts agree with all three supplied metric tables; training and validation counts are derived from the specified window rule, not reported epoch logs.

Season	Interval (end exclusive)	Hours	Train	Validation	Test
Spring	2025-03-01 to 2025-06-01	2208	1757	219	219
Summer	2025-06-01 to 2025-09-01	2208	1757	219	219
Autumn	2025-09-01 to 2025-12-01	2184	1738	216	217
Winter	2024-12-01 to 2025-03-01	2160	1719	214	214

There are 869 reported test forecasts per horizon across seasons, or 2607 forecast–target pairs per model when horizons are counted separately. Overlapping multihorizon windows do not make these independent observations. Equal sample counts and season labels support comparison but are not, by themselves, proof of identical target timestamps and values.

4.2 Parameter setting

Table 4 gives the retained source-code defaults, distinguishing them from unavailable executed-run artifacts. The baseline source files were protected: copies were checked for byte identity. New-data date selection, schema conversion and reporting are separate adapters. No baseline architecture, optimizer or residual scaler is silently replaced by the wider spatial project's different settings.

Finalized versus searched parameters. L's K values and the search grids are known, but the reported new-data tables do not include selected alpha values or L.1's selected K values. They are therefore not filled using historical winners from a different dataset. The actual selected-pair files must establish those outcomes. Likewise, best epoch, epochs executed, final feature widths, parameter counts and run time are not inferred from the accuracy table.

Execution and evidence status. The comparison wrappers default to CPU because the protected baseline validation routine constructs a CPU tensor without moving it to CUDA. The new spatial

Table 4: Documented comparison configuration. These are source defaults; the complete run configuration files were not supplied with the scores.

Setting	Value or procedure
Target and outputs	Station 47; horizons 1, 2 and 3 hours
Seasonal split	80% / 10% / 10%, with boundary-window removal
Meteorological inputs	Spring/autumn: WS, ET; summer: WS, ET, AT, RH; winter: WS
L mode counts	11 / 10 / 11 / 10, seasonal order above
L.1 candidate K; alpha grid	K=2–12; alpha in {25,100,500,2000}
VMD step, tolerance, initialization	0; 10^{-7}; uniform; 500-iteration package limit
Entropy and EWT	$m = 2,\ r = 0.2\,\mathrm{std},\ \mathrm{SE} > 0.4$; eight EWT modes
Feature bank and input scaling	All VMD + EWT + selected weather + raw WS; train-fitted StandardScaler
Ridge penalty and inputs	0.1; own Brentwood features only
Residual transform	Training-fitted MinMaxScaler
Window; decoder label length	7; 4
Hidden dimension; attention heads	64; 8
Encoder / decoder blocks	2 / 1; encoder distillation between blocks
KAN hidden width; grid; spline order	20; 5; cubic
Dropout; attention factor	0; 5
Optimizer; clipping	Adam, lr 0.001; gradient norm cap 1
Schedule	Cosine decay to lr/20 over configured epochs
Epoch limit; patience; batch size	300; 30 non-improving epochs; 32
Training sample order; seed	No shuffle; default seed 42
Graph default	Geographic, symmetrized 3-nearest support, Gaussian weights, self loops, row normalization

adapter supports its own input-device transfer, but a hardware/software match for the reported executions is not established. No GPU speed, memory or energy result is claimed.

The evidence available for this draft is the implementation snapshot, the copied-source integrity checks and the user's three aggregate metric tables. The actual prediction CSVs, checkpoints, run protocols, selected parameter records and seasonal exclusion manifests are not supplied. The local collector is designed to check matching targets and settings when those artifacts are available; the reported-score table alone cannot substitute for that verification. A remote neighbor-handling change was reported after the local snapshot, and its exact executed code should be archived before publication.

4.3 Evaluation metrics

For seasonal test origins $\mathcal{Q}_s$, $n_s = |\mathcal{Q}_s|$, and $e_{o,h} = \widehat{y}_{o,h} - y_{o,h}$,

$$\mathrm{RMSE}_{s,h} = \sqrt{\frac{1}{n_s} \sum_{o \in \mathcal{Q}_s} e_{o,h}^2}, \tag{29}$$

$$\mathrm{MAE}_{s,h} = \frac{1}{n_s} \sum_{o \in \mathcal{Q}_s} |e_{o,h}|, \tag{30}$$

$$\mathrm{MAPE}_{s,h} = \frac{100}{n_s} \sum_{o \in \mathcal{Q}_s} \left| \frac{e_{o,h}}{y_{o,h}} \right|. \tag{31}$$

RMSE and MAE are in $\mathrm{m\,s^{-1}}$; MAPE is percent. Scoring is for Brentwood only and after inverse residual transformation and Ridge addition. No across-station averaging is used in this comparison.

The protected metric rejects a horizon containing exact-zero target wind. A separately labelled external diagnostic excludes exact zeros, preserving their count. All supplied rows report zero such targets, so strict MAPE and that diagnostic coincide. The different $0.1\,\mathrm{m\,s^{-1}}$ MAPE threshold used by the wider multistation runner is not applied here.

Spatial relative improvement against baseline B is

$$I_{s,h}^{(m)}(\mathrm{S:B}) = 100\,\frac{E_{s,h}^{(m)}(\mathrm{B}) - E_{s,h}^{(m)}(\mathrm{S})}{E_{s,h}^{(m)}(\mathrm{B})}. \tag{32}$$

Positive values indicate lower spatial error. Winners use supplied unrounded values; tables round only for display. A case is one season–horizon pair; a metric comparison is one case and one of the three error measures. MAPE and its identical nonzero diagnostic are not counted twice.

5 Comparison results and discussion

5.1 Single-model comparison scope

The three evaluated systems combine Ridge and a neural residual model. L is the fixed-K prediction-selected baseline; L.1 is the local hybrid selection baseline; Spatial L.1 adds the target-specific graph layer. The old-period publication scores and earlier Brentwood runs are omitted from the new-data winner table because their evaluation periods differ.

No standalone Ridge, standalone KANInformer, persistence or neighbor-Ridge scores were supplied for these particular seasonal runs. Results from the separate whole-network stage cannot be substituted: they use different processing, partitions and station averaging. Accordingly, the observed combined errors cannot establish how much the neural residual improves upon local Ridge in this experiment.

5.2 Primary-decomposition diagnostics

The methodology specifies reconstruction scoring, entropy grouping, frequency ordering and EWT synthesis. However, the new-data selection and decomposition reports were not supplied. Actual seasonal RER values, selected high-entropy groups, alpha/K winners and EWT reconstruction errors are not invented. A previous dataset's K=12 outcome is not evidence that this new dataset selected K=12.

A reproducible diagnostic record should report candidate RER curves, four alpha-finalist validation scores, the selected pair, whether K lies at its search boundary, entropy values, EWT synthesis residuals and per-node feature widths. Such records would distinguish better signal reconstruction from useful target prediction. The available aggregate test scores establish neither decomposition mechanism nor physical interpretability of individual modes.

5.3 Forecasting performance after dual decomposition

Table 5 reports the complete supplied comparison. Every season–horizon group has the same reported test count across models. All 108 metric entries are transcribed from reported scores; they have not been independently reconstructed from prediction files. Spatial figures were supplied with fewer decimal places than the two local tables. That difference is preserved in the evidence snapshot and rounding is used only for presentation.

For all three systems, the spring and summer errors generally increase with horizon, whereas other seasons can show different relative patterns. The smallest error does not always belong to the same architecture across metrics. For example, summer horizon two favors L in RMSE/MAPE but L.1 in MAE. A single categorical label for that case would hide this distinction.

Table 5: New-data Brentwood errors for L, L.1 and Spatial L.1 (S). RMSE/MAE are in $\mathrm{m\,s^{-1}}$, MAPE in percent; bold is the lowest error within each row and metric group.

		RMSE			MAE			MAPE (%)		
Season	Hours	L	L.1	S	L	L.1	S	L	L.1	S
Spring	1	0.010664	**0.010257**	0.010466	0.008682	**0.008202**	0.008485	0.567	**0.536**	0.575
Spring	2	0.014505	0.014279	**0.013817**	0.012188	0.011989	**0.011546**	0.804	0.788	**0.767**
Spring	3	0.027839	0.027216	**0.026147**	0.023496	0.022899	**0.021876**	1.552	1.509	**1.463**
Summer	1	0.013240	**0.012260**	0.012280	0.009409	**0.008470**	0.008481	0.623	0.558	**0.557**
Summer	2	**0.021914**	0.021960	0.022213	0.015080	**0.014995**	0.015168	**1.009**	1.031	1.035
Summer	3	**0.038127**	0.038503	0.038894	0.026292	**0.025948**	0.026251	1.764	**1.762**	1.773
Autumn	1	0.004493	**0.004272**	0.004635	0.003592	**0.003298**	0.003720	0.345	**0.318**	0.360
Autumn	2	0.005055	0.005147	**0.004772**	0.004060	0.004113	**0.003753**	0.398	0.393	**0.369**
Autumn	3	0.010045	**0.009210**	0.009267	0.008093	**0.007247**	0.007325	0.789	**0.697**	0.715
Winter	1	0.008310	**0.007809**	0.008070	0.006639	**0.006188**	0.006315	0.461	**0.431**	0.437
Winter	2	**0.009382**	0.010257	0.009760	**0.007204**	0.008178	0.007542	**0.510**	0.579	0.522
Winter	3	0.016329	**0.012786**	0.013711	0.012702	**0.009911**	0.010545	0.893	**0.696**	0.743

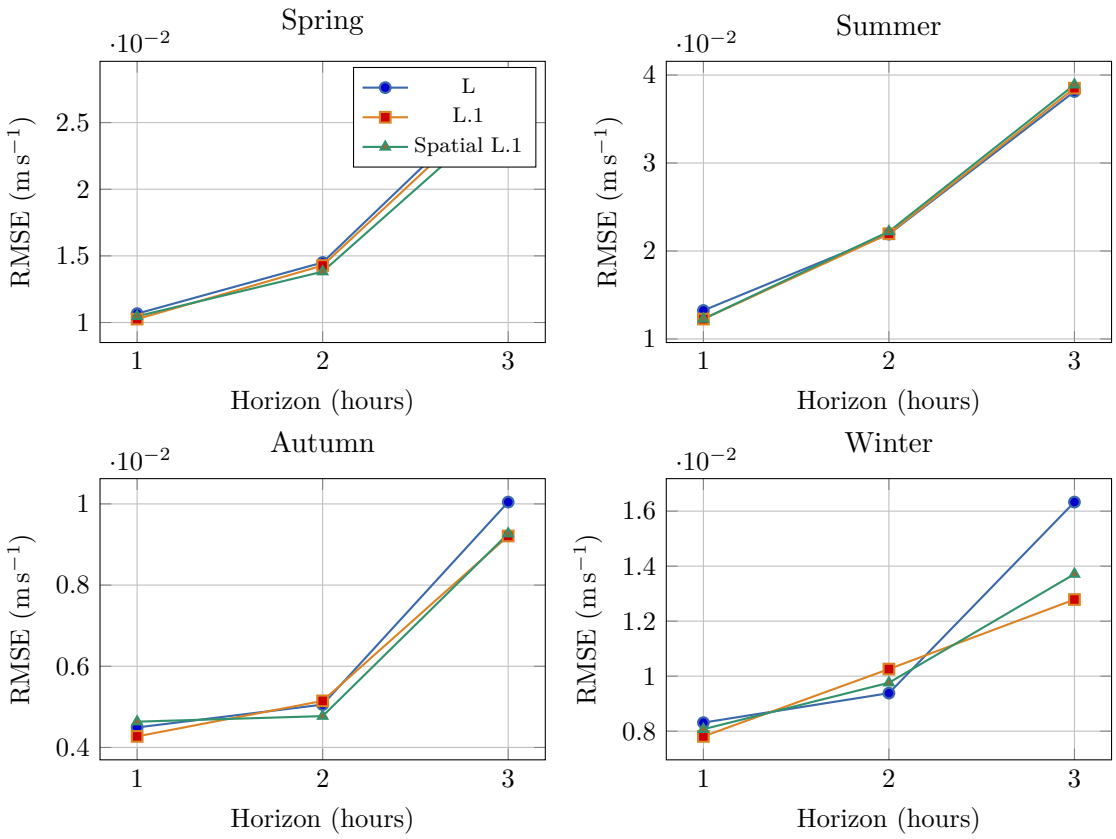


Figure 4: Reported seasonal RMSE versus forecast horizon. These are plots of aggregate metric values, not reconstructed wind-speed prediction curves.

Figure 4 displays the mixed horizon-dependent outcome. The spatial curve is below L.1 for spring horizons two and three, autumn horizon two and winter horizon two, but not for the remaining RMSE comparisons. The close curves in several cases indicate small absolute differences, and line placement is not evidence of statistical significance.

5.4 Comparison with the local hybrid reference

The direct test of the added spatial pathway is Spatial L.1 versus L.1. Table 6 uses Equation (32); negative values denote deterioration. The spatial model improves all three metrics in four cases: spring horizons two and three, autumn horizon two, and winter horizon two. It also slightly improves summer horizon-one MAPE, while its RMSE and MAE there are slightly worse.

Table 6: Spatial L.1 relative improvement over L.1 (%). Positive values are improvements, negative values are increases in error.

Season	Hours	RMSE	MAE	MAPE
Spring	1	-2.04	-3.46	-7.22
Spring	2	3.23	3.69	2.65
Spring	3	3.93	4.47	3.04
Summer	1	-0.16	-0.13	0.16
Summer	2	-1.15	-1.15	-0.39
Summer	3	-1.02	-1.17	-0.61
Autumn	1	-8.50	-12.77	-13.15
Autumn	2	7.29	8.76	6.13
Autumn	3	-0.62	-1.07	-2.57
Winter	1	-3.34	-2.05	-1.39
Winter	2	4.85	7.78	9.73
Winter	3	-7.23	-6.39	-6.77

The largest spatial RMSE improvement over L.1 is 7.29% at autumn horizon two. Winter horizon two improves RMSE by 4.85%, MAE by 7.78% and MAPE by 9.73%. That winter improvement is relative to L.1; L still has the lowest three errors in that case. Thus a spatial improvement over its direct parent need not be the minimum among all three systems.

Conversely, autumn horizon one worsens by 8.50% in RMSE, 12.77% in MAE and 13.15% in MAPE relative to L.1. Winter horizon three worsens by 7.23% in RMSE. These counterexamples prevent a conclusion of consistent spatial benefit. They do not identify the cause without the final graph, prediction residuals and controlled ablations.

Table 7: Per-metric winner counts across twelve cases, and direct Spatial L.1 versus L.1 outcomes.

	Three-system minimum			S versus L.1	
Metric	L	L.1	S	Improves	Worsens
RMSE	3	6	3	4	8
MAE	1	8	3	4	8
MAPE	2	6	4	5	7
Total	6	20	10	13	23

L.1 achieves 20 of 36 reported minima, Spatial L.1 achieves 10 and L achieves 6. L.1 wins all three metrics in five cases; Spatial L.1 does so in three cases and L in one; the remaining three have mixed winners. Directly against L.1, Spatial L.1 improves 13 and worsens 23 metrics. The overlapping horizons and correlated error measures mean these are descriptive counts, not 36 independent statistical trials.

Equal-weight case means. To complement wins, define

$$\overline{E}^{(m)}(M) = \frac{1}{12} \sum_{s\in\{\text{spring,summer,autumn,winter}\}} \sum_{h=1}^{3} E_{s,h}^{(m)}(M). \tag{33}$$

This is an average of reported case metrics, not an RMSE recomputed from pooled residuals. It gives every season–horizon case equal weight despite slightly different sample counts.

Table 8: Equal-weight mean errors across twelve cases. Bold marks the lowest mean. The final column is S's relative improvement over L.1.

Metric	L	L.1	S	S versus L.1 (%)
RMSE	0.014991816	**0.014496297**	0.014502534	-0.043
MAE	0.011453139	0.010953198	**0.010917171**	0.329
MAPE (%)	0.809586	**0.774862**	0.776390	-0.197

The spatial extension reduces mean MAE by approximately 0.329%, but its mean RMSE is 0.043% higher and mean MAPE is 0.197% higher than L.1. L.1 therefore wins more individual comparisons, while Spatial L.1 has a marginally lower equal-weight mean MAE. These summaries answer different questions and neither justifies saying one model dominates every criterion.

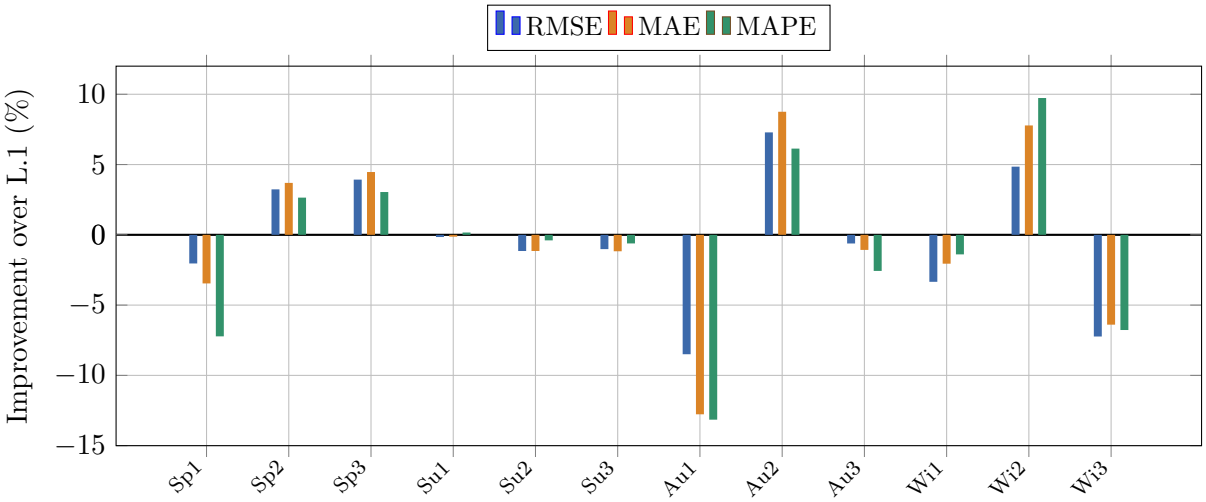


Figure 5: Spatial improvement relative to L.1. Labels combine season (Sp, Su, Au, Wi) and horizon. Bars above zero indicate improvement; bars below zero indicate deterioration.

Figure 5 makes selective transfer visible: the strongest benefits and deteriorations occur in different seasonal/horizon cases. It is an aggregate error comparison, not a map of which station caused a change and not a visualization of learned physical wind transport.

5.5 Prediction diagnostics and interpretability scope

The available tables contain nonnegative aggregate errors and sample counts, not individual signed residuals. They cannot recover forecast bias, residual autocorrelation, event-specific failures, prediction interval coverage, correlation of forecast errors across stations, or a measured lead–lag propagation pattern. Observed-versus-predicted curves, scatter diagrams, SHAP values and attention interpretations are therefore not fabricated.

The source architecture establishes where spatial information can enter; it does not establish which neighbor supplied useful information in the reported run. Required supporting artifacts include the final target-row weights, retained station order, projected feature dimensions, per-station

parameter reports and prediction residuals. Fixed edge weight magnitude is not a learned causal importance score. Neighbor projection weights can also interact with the feature scaler and with downstream normalization.

5.6 Discussion

(1) Why the spatial idea was reasonable. Local temporal histories omit contemporaneous conditions observed elsewhere. A fixed spatial graph is a simple way to test whether those observations provide useful context before replacing the entire temporal backbone with a larger architecture. The target-only design also preserves the practical question: improving Brentwood rather than improving an average across unrelated station outputs.

(2) What the reported results establish. Spatial context can coincide with lower errors for specific cases, particularly spring horizons two/three and autumn horizon two. It also improves all winter horizon-two errors against L.1, though not against L. The overall win count favors L.1 and most direct comparisons worsen. The defensible finding is selective benefit in a mixed result, not confirmation that adding stations necessarily improves accuracy.

(3) Possible explanations requiring tests. A strong local Ridge-plus-neural representation may already contain most usable information in this retrospective setting. Geographic neighbors may be poorly timed or statistically redundant. Eligibility changes alter the network between seasons. Learned projections and LayerNorm may modify useful own-station features, even when neighbor information is weak. These are hypotheses consistent with the architecture, not mechanisms demonstrated by the aggregate table.

(4) Attribution and fair controls. The key baseline is L.1, because L versus Spatial L.1 also changes K selection. Even L.1 versus Spatial L.1 changes both information and adapter capacity. A self-only adapter with the same nonlinear transform and normalization would test capacity effects; shuffled-neighbor and zero-neighbor controls would test the information content. A local Ridge-only forecast would quantify the incremental neural contribution. None of these extra scores is available for the reported seasonal runs.

(5) Leakage and the scale of the errors. Full-season decomposition exposes each representation to future data. Full-season outlier masking and interpolation add further dependence and can smooth evaluated targets. The low reported errors should not be interpreted as evidence that an operational system can forecast unknown wind with equivalent precision. A causal counterpart requires chronological cleaning, past-only decomposition and validation-based parameter selection without full-season test information. Preserving the leakage protocol allows an internal retrospective comparison; it does not remove the need for that separate study.

(6) Stability, cost and deployment. Multiple seeds and rolling-origin evaluations are needed to distinguish small differences from optimization variability. ProbSparse key sampling is stochastic in the retained backbone, so a default seed is not a cross-device bitwise reproducibility guarantee. Neighbor-wise hybrid search adds substantial CPU work and extra failure points. No runtime or deployment efficiency measurement was supplied, and no claim of speed improvement follows from the accuracy table.

6 Conclusion

6.1 Theoretical contributions

This study develops and documents a target-specific spatial extension of a decomposition-assisted Ridge–KANInformer implementation. The three systems distinguish fixed-K prediction-guided selection (L), hybrid reconstruction/prediction selection (L.1), and graph-enriched residual prediction (Spatial L.1). The temporal backbone and Brentwood-only Ridge path are preserved, while neighbor features are projected to the target feature dimension and aggregated using a fixed graph row.

The reported 2025 Brentwood comparison does not show uniform superiority of the spatial extension. L.1 achieves 20 of 36 metric minima, Spatial L.1 achieves 10 and L achieves 6. Against L.1, the spatial model improves 13 metrics and worsens 23, with complete three-metric gains in four season–horizon cases. Equal-weight mean MAE improves slightly, while mean RMSE and MAPE are slightly higher. The contribution is therefore a specified architecture and a transparent mixed-result comparison, rather than a theoretical proof or an unconditional accuracy claim.

A practical implication follows: adding neighboring data should be treated as an empirical hypothesis, assessed against strong local baselines on common targets. Spatial autocorrelation, model complexity and additional stations do not by themselves establish forecast value.

6.2 Limitation and further research

The main limitations are the inherited full-season leakage, processed rather than necessarily untouched observed targets, a single reported run per architecture, absent prediction-level records and incomplete executed-version metadata. The reported neighbor-exclusion modification must be archived with the actual station sets and adjacency. Selected alpha/K values, epoch histories, hardware and overrides cannot be recovered from aggregate errors.

Future work should proceed in the following order:

1. **Complete the evidence archive.** Save executed code versions, source/data hashes, selected pairs, neighbor exclusions, final graph, scalers, predictions and checkpoints. Verify identical Brentwood target records before attributing differences to architecture.

2. **Measure repeatability.** Repeat matched configurations over several seeds and chronological evaluation folds. Use time-aware uncertainty analysis on prediction errors rather than interpreting metric-win counts as independent significance tests.

3. **Separate representation from spatial effects.** Compare fixed-alpha reconstruction selection, the present hybrid selection and prediction selection under the same input period and budget. Add local Ridge-only and neural-only controls.

4. **Test the adapter itself.** Include self-only, zero-neighbor and shuffled-neighbor adapters; vary geographic support and compare training-derived statistical graphs. Select these choices on validation, not on the reported test table.

5. **Evaluate past-only information.** Use chronological imputation, past-only or rolling-context decomposition and immutable observed-target masks. Compare its accuracy and cost with a no-decomposition baseline.

6. **Extend only when justified.** Investigate lag-aware or wind-direction-informed edges, learned attention and richer spatiotemporal architectures when simpler controls show reliable incremental spatial information. Such models would require new experiments rather than being inferred from this result.

The present evidence supports investigating spatial context further, while retaining L.1 as the stronger metric-win baseline for these reported seasonal results.

CRediT authorship contribution statement

Author identities and individual contributions are to be supplied by the authors before submission.

Declaration of competing interest

The authors must supply their competing-interest declaration before submission; no declaration is inferred from the code or score tables.

Acknowledgments

Funding, institutional support and acknowledgments are to be completed by the authors. The meteorological data source is California CIMIS.

Data and code availability

The implementation is organized as a separate comparison project with L, L.1 and Spatial L.1 runners. Protected local source copies are kept separate from new-data adapters and the graph module. Individual experiment outputs belong under their own result folders; the combined report is generated after matched-record checks when full artifacts are available.

The evidence snapshot for this manuscript contains three reported metric tables with twelve rows each for station 47 and year 2025. Spatial values retain the lower numerical precision supplied. These tables are available locally under `compare/results/reported_2025/`; their provenance explicitly states that prediction timestamps and matching executed configurations were not independently verified from the supplied artifacts. No new model training was performed to draft the manuscript.

The remote seasonal-neighbor exclusion fix and its run manifests should be deposited with the exact executed source before public release. A source-code-only ZIP does not contain the station datasets, requirements, checkpoints or the complete experiment evidence. Public archive identifiers, licensing and access permissions must be supplied by the authors. This manuscript contains all its numerical tables, diagram definitions and plotted coordinates directly in the LaTeX source; no external graphics or bibliography file is required.

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